

Instructions:

- **Due:** February 18, 11:59PM . Upload one **single PDF** to Blackboard containing your complete submission and upload all associated L^AT_EX source files.
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page.
- Label your PDF file exactly as

ENME408_S26_HW03_YourLastName.pdf

- All work must be typeset in L^AT_EX. Handwritten or scanned submissions will not be accepted.
- Failure to follow these submission and formatting guidelines may result in a substantial point deduction or a score of zero.

Problem 1. Let x, y, z denote the vertices of an equilateral triangle. Show that $\vec{r}_{y/x} = \vec{r}_{y/z} + \vec{r}_{z/x}$. **Hint:** Define a frame and write the three physical vectors in terms of the frame unit vectors.

Problem 2. Let frames F_A, F_B, F_C be such that

$$\hat{i}_B = \frac{\sqrt{2}}{2}\hat{i}_A + \frac{\sqrt{2}}{2}\hat{j}_A, \quad \hat{j}_B = -\frac{\sqrt{2}}{2}\hat{i}_A + \frac{\sqrt{2}}{2}\hat{j}_A, \quad \hat{k}_B = \hat{k}_A, \quad (1)$$

$$\hat{i}_C = \hat{i}_B, \quad \hat{j}_C = \frac{1}{2}\hat{j}_B + \frac{\sqrt{3}}{2}\hat{k}_B, \quad \hat{k}_C = -\frac{\sqrt{3}}{2}\hat{j}_B + \frac{1}{2}\hat{k}_B. \quad (2)$$

Obtain $\mathcal{O}_{B/A}, \mathcal{O}_{C/B}$, and $\mathcal{O}_{C/A}$.

Problem 3. Let frames F_A, F_B, F_C, F_D be defined by the sequence of rotations

$$F_A \xrightarrow{3} F_B \xrightarrow{2} F_C \xrightarrow{1} F_D \quad (3)$$

and let $\vec{x} = \ell_1\hat{i}_D + \ell_2\hat{j}_D + \ell_3\hat{k}_D$. Obtain $\vec{x}|_A$ in terms of ℓ_1, ℓ_2, ℓ_3 , $e_1 \triangleq \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $e_2 \triangleq \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $e_3 \triangleq \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and the Euler orientation matrices.

Problem 4. Let frames F_A, F_B, F_C be such that

$$F_A \xrightarrow{2} F_B \xrightarrow{1} F_C$$

where $\theta(t) = 0.6 \sin(2\pi t)$ and $\phi(t) = 0.3 \sin(9\pi t)$, for all $t \geq 0$, which implies that

$$\vec{\omega}_{C/A} = \vec{\omega}_{C/B} + \vec{\omega}_{B/A} = \dot{\phi}\hat{i}_C + \dot{\theta}\hat{j}_B.$$

Obtain the following:

1. $\vec{\omega}_{C/A}|_C$ at time $t = 0.01$ s.
2. $\vec{\alpha}_{C/A}|_C$ at time $t = 0.01$ s.

Hint: Note that $\vec{\alpha}_{C/A} = \frac{A\bullet}{\vec{\omega}_{C/A}} = \frac{C\bullet}{\vec{\omega}_{C/A}}$. Numerical solution:

$$\vec{\omega}_{C/A}(0.01)\Big|_C = \begin{bmatrix} 8.1455 \\ 3.7493 \\ -0.3145 \end{bmatrix} \text{ rad/s},$$

$$\vec{\alpha}_{C/A}(0.01)\Big|_C = \begin{bmatrix} -66.9108 \\ -4.0442 \\ -30.4156 \end{bmatrix} \text{ rad/s}^2.$$

Problem 5. Consider the rotating-rods mechanism shown in Figure 1. A rod of length l_1 is connected by a link to a vertical shaft that is fixed to the ground at point o . The rod is perpendicular to the shaft and is free to rotate about it; point a is fixed at the free end of this rod. A second rod of length l_2 is connected to the first rod by a pin joint at point a ; point b is fixed at the free end of the second rod. Let frame F_A be fixed in space as shown in Figure 1. Let frames F_B and F_C be rigidly attached to line segments oa and ab , respectively, as shown in Figure 1. Define ϕ as the angle from $\hat{i}_A$ to $\hat{i}_B$ measured about $\hat{j}_A$, and define θ as the angle from $\hat{i}_B$ to $\hat{i}_C$ measured about $\hat{k}_B$.

The frames are thus related by

$$F_A \xrightarrow{\frac{\phi}{2}} F_B \xrightarrow{\frac{\theta}{3}} F_C$$

which implies that

$$\vec{\omega}_{B/A} = \dot{\phi} \hat{j}_A, \quad \vec{\omega}_{C/B} = \dot{\theta} \hat{k}_B$$

Assume that both rods are rotating at constant angular rates, that is, $\dot{\phi}$ and $\dot{\theta}$ are constant.

Obtain an expression in terms of $l_1, l_2, \phi, \dot{\phi}, \theta, \dot{\theta}$ for

1. $\vec{v}_{b/o/A}\Big|_A = \frac{A\bullet}{\vec{r}_{b/o}}\Big|_A$.
2. $\vec{a}_{b/o/A}\Big|_A = \frac{A\bullet}{\vec{v}_{b/o/A}}\Big|_A = \frac{A\bullet\bullet}{\vec{r}_{b/o}}\Big|_A$.

Hint: Note that $\vec{v}_{b/o/A} = \vec{v}_{b/a/A} + \vec{v}_{a/o/A}$, and that $\vec{a}_{b/o/A} = \frac{A\bullet}{\vec{v}_{b/o/A}}$.

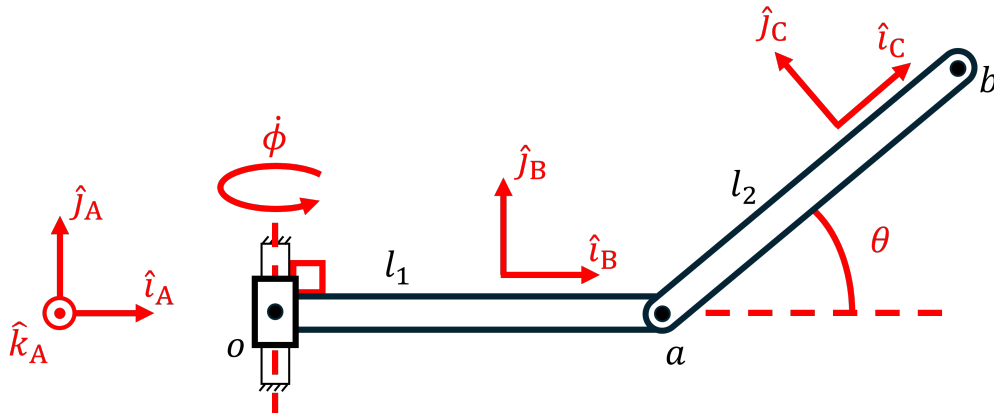


Figure 1: Rotating-rods mechanism.

Extra Credits Problem 1. [30 points.] Consider the slider-crank mechanism shown in Figure 2. The crank is a rigid disk of radius r that rotates about a fixed pin joint at point o . The connecting rod has constant length l and is attached to the disk by a pin joint at point a . The slider is constrained to move only horizontally and is connected to the connecting rod by a pin joint at point b . Let frame F_A be fixed in space as shown in Figure 2. Note that the line segment ob is always aligned with the unit vector $\hat{i}_A$. Let frames F_B and F_C be rigidly attached to line segments oa and ab , respectively, as illustrated in Figure 2. Define ϕ as the angle between oa and ob , and define θ as the angle between ba and bo . The frames are thus related by

$$F_A \xrightarrow{\frac{\phi}{3}} F_C, \quad F_A \xrightarrow{\frac{\theta}{3}} F_C,$$

which implies that

$$\vec{\omega}_{B/A} = \dot{\phi} \hat{k}_A, \quad \vec{\omega}_{C/A} = \dot{\theta} \hat{k}_A.$$

Note that in the configuration of the slider-crank mechanism shown in Figure 2, ϕ is a positive number, whereas θ is a negative number. Assume that the disk is rotating at a constant angular rate, that is, $\dot{\phi}$ is a constant. Let $\vec{r}_{b/o} = x \hat{i}_A$, where $x \in \mathbb{R}$. Obtain an expression for $\dot{\theta}$ and $\dot{x}$ in terms of r, l, ϕ, θ , and $\dot{\phi}$.

Hint: Compute both sides of the equation $\vec{v}_{b/o/A} = \vec{v}_{b/a/A} + \vec{v}_{a/o/A}$ and compare coefficients of frame vectors.

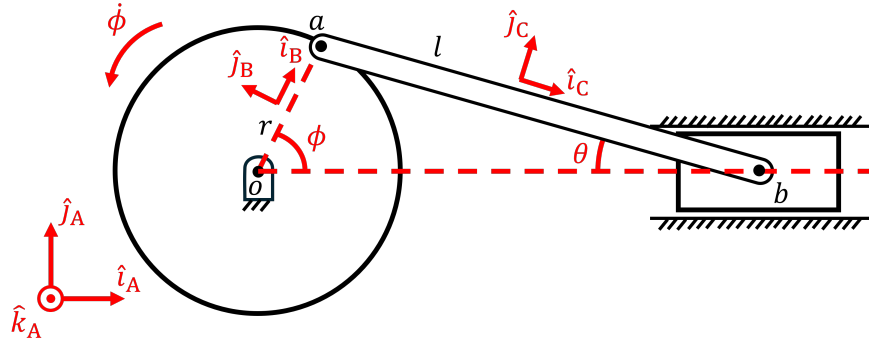


Figure 2: Slider-crank mechanism